

SDI-12 Environmental Sensor - Program Overview

This document describes the firmware in the main/ folder: an Arduino-based environment sensor that reads BME280 and BH1750, acts as an SDI-12 slave, logs data to SD card with RTC timestamps, and shows a real-time TFT dashboard. Generated for the REAL ONE project.

1. What the Program Does

- Read sensors: temperature, humidity, pressure (BME280), light level (BH1750) over I2C.
- SDI-12 slave: respond to a data logger on Serial1 at address 0 (configurable).
- Local logging: append CSV rows to datalog.csv on SD card every 60 seconds.
- Manual control: pushbuttons for manual log entry and clearing the log file.
- TFT dashboard: ST7735 display showing live sensor values.

2. Recommended Reading Order

#	File	Purpose
1	main.ino	Entry point: setup() init order and loop() schedule.
2	HardwareConfig.h	Pin numbers and timing constants.
3	SensorReading.h / .cpp	SensorData struct, readSensors(), getSensorData().
4	sdi12.h / .cpp	SDI-12 protocol and command parser.
5	DataLogger.h / .cpp	RTC, SD card, buttons, CSV logging.
6	Dashboard.h / .cpp	ST7735 TFT user interface.

3. Project File Structure (main/)

main.ino	Orchestrator (setup + loop)
HardwareConfig.h	Pins and intervals
SensorReading.h/cpp	BME280 + BH1750 driver layer
sdi12.h/cpp	SDI-12 slave on Serial1
DataLogger.h/cpp	RTC, SD card, buttons, CSV
Dashboard.h/cpp	ST7735 TFT display

4. main.ino - Program Flow

setup() - runs once at boot

- Serial.begin(9600) - USB debug port.
- sensorsInit() - I2C, BME280, BH1750.
- dataLoggerInit() - RTC, SD card, buttons.
- dashboardInit() - TFT display.
- sdi12Init(MY_ADDRESS, DIRO_PIN) - Serial1 @ 1200 7E1, direction pin.
- readSensors() - first sample so buffer is not empty.

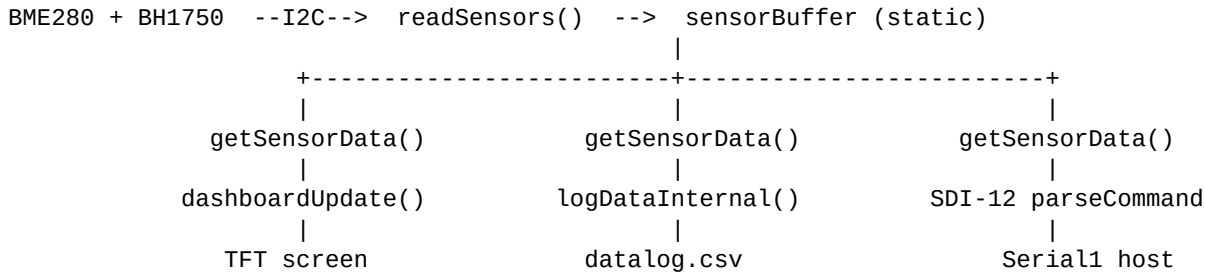
- Every 500 ms (kSensorPollMs): readSensors() updates shared buffer.
- sdi12Handle() - process incoming SDI-12 commands (non-blocking).
- dataLoggerUpdate() - 60 s auto-log, button checks.
- dashboardUpdate() - refresh TFT about every 1 second.

kSensorPollMs (500): minimum interval between background sensor reads.

lastSensorPollMs: millis() timestamp of the last readSensors() call.

These use a non-blocking timer so loop() never calls delay() for polling.

5. Data Flow



SensorReading.cpp owns the buffer. All other modules call `readSensors()` when they need fresh data, or rely on the 500 ms background poll in `main.ino`.

6. Module Reference

SensorReading

- `SensorData`: temperature, humidity, pressure, lux, ready flag.
- `sensorsInit()`, `readSensors()`, `getSensorData()`.
- `isBme280Ok()`, `isBh1750Ok()`, `getParameterCount()` for SDI-12.

- `Serial1` @ 1200 baud, 7 data bits, even parity, 1 stop bit.
- `DIRO_PIN`: TX/RX direction for line driver (pin 8 in `main.ino`; pin 7 is `TFT_DC`).
- Commands assembled until `!` then parsed in `parseCommand()`.
- Debug `[RX]/[TX]` printed on USB Serial, not on SDI-12 bus.

- RTC DS1307 for timestamps.
- SD card via soft SPI (`SdFat`); file: `dataLog.csv`.
- CSV columns: Timestamp, Temperature_C, Humidity_pct, Pressure_hPa, Lux, Source.
- Source: Periodic (60 s) or Manual (button pin 2).
- Button pin 3: delete log file and recreate CSV header.

- Adafruit ST7735 TFT; `INTR_BLACKTAB`, rotation 1.
- Static labels drawn once in `dashboardInit()`; values updated in `dashboardUpdate()`.
- Refresh interval: `kDashboardRefreshMs` (1000 ms) in `HardwareConfig.h`.

7. Hardware Pins (HardwareConfig.h + main.ino)

Signal	Pin(s)	Notes
SD CS	A3	SD card chip select (soft SPI)
SD MISO / MOSI / SCK	12 / 11 / 13	Soft SPI for SD card
Btn manual log	2	Active LOW, INPUT_PULLUP
Btn clear SD	3	Active LOW, INPUT_PULLUP
TFT CS / DC / RST	10 / 7 / 6	ST7735 control pins
TFT MOSI / SCLK	11 / 13	Hardware SPI (shared with SD soft SPI)
DIRO_PIN	8	SDI-12 direction (<code>main.ino</code> ; not pin 7)
I2C	SDA/SCL	BME280 @ 0x76, BH1750, RTC DS1307
Serial1	TX/RX	SDI-12 data line to host

8. SDI-12 Commands (address 0)

Command	Description
?	Address query - returns sensor address
0M	Measure - returns 0003 + parameter count, reads sensors
0D1	BME280 data: temp, humidity, pressure
0D2	BH1750 data: lux
0R0	All data: temp, humidity, pressure, lux
0An	Change address to character n

9. Required Arduino Libraries

- Adafruit BME280 Library + Adafruit Unified Sensor
- BH1750
- Adafruit GFX Library + Adafruit ST7735 and ST7789 Library
- RTCLib (Adafruit)
- SdFat (Bill Greiman v2+)
- Bounce2

10. Debugging Tips

- Open USB Serial Monitor at 9600 baud for [RX], [TX], [Log], [Init] messages.
- If TFT is blank: check wiring, INTR_BLACKTAB vs GREENTAB, I2C sensor init.
- If SDI-12 fails: verify DIRO_PIN wiring, Serial1 pins, MY_ADDRESS matches host.
- If RTC time is wrong: replace battery or run rtc.adjust() once in DataLogger.cpp.
- If SD fails: check CS on A3 and soft SPI pins 11/12/13.

Project path: REAL ONE/main/ | Document auto-generated from firmware structure.